

Module-End -1

Module End Assignment:

“Data Cleaning, Analysis & Visualization with Excel”

STEP-1

The 4 datasets are given in the Ecommerce_sales_dataset.

They are .,Customer_Dim,Product_Dim,Store_Dim and Sales_Fact.

The datas in the tables are converted into the TABLE format using CTRL+T option.

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	sheena bi	55	Male	Donaldson	North Car	USA	
CUST1001	Eugene Ne	26	Female	Phillipsmouth	North Dak	USA	Platinum
CUST1002	dr. Benjarr	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	jeremy har	59	Female	Thornastown	Ohio	USA	Bronze
CUST1004	Caroline C	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine V	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber lac	29	Female	Deborahborough	South Dak	USA	Platinum
CUST1007	Daniel Goy	29	Female	West Pamelafurt	West Virgi	USA	Bronze
CUST1008	Melissa He	54	Female	Maynardton	New Hamy	USA	Gold
CUST1009	Ms. Maria	64	Female	Melissashire	Pennsylvan	USA	Platinum
CUST1010	Dominic Pi	53	Male	Port Michaelburgh	Pennsylvan	USA	Bronze
CUST1011	Brian Lope	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth L.	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan He	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Col	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alva	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	Iohn Scott	24	Female	South Christopher	Pennsylvan	USA	Bronze
CUST1017	Adam Cru	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda M	43	Male	South William	Kansas	USA	
CUST1019	Michael Tr	46	Female	Gonzalezfort	New Jerse	USA	Platinum
CUST1020	Bridget All	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martin	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	paul fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan t	52	Female	Molinebury	Oregon	USA	Silver
CUST1024	Heather Gi	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Toy	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Can	18	Female	East Michellermouth	Missouri	USA	Gold
CUST1027	Leslie Johr	31	Male	Mollyshire	Utah	USA	
CUST1028	Joshua Ryc	27	Female	New David	New Jerse	USA	Platinum
CUST1029	David Arnc	54	Female	East Jennifer	California	USA	Silver
CUST1030	Jessica Acc	64	Female	Amnybury	Pennsylvan	USA	Bronze
CUST1031	Seash Uva	41	Male	Johnsborough	Oregon	USA	Silver

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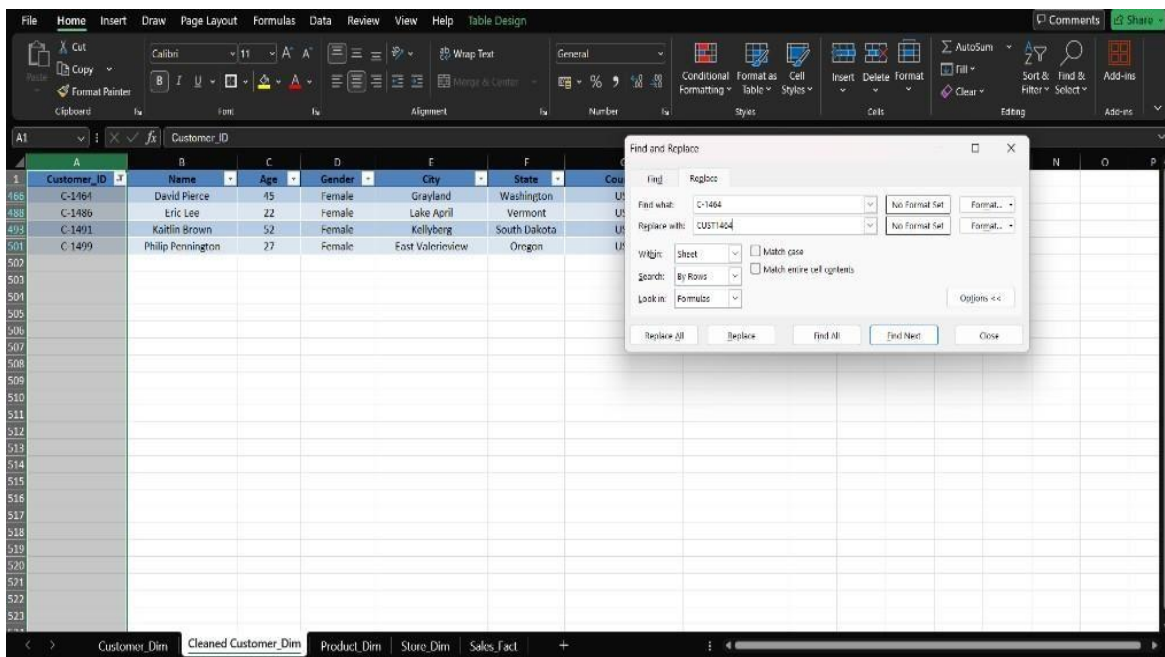
Data entry ,import &organisation using tables are done.

STEP-2

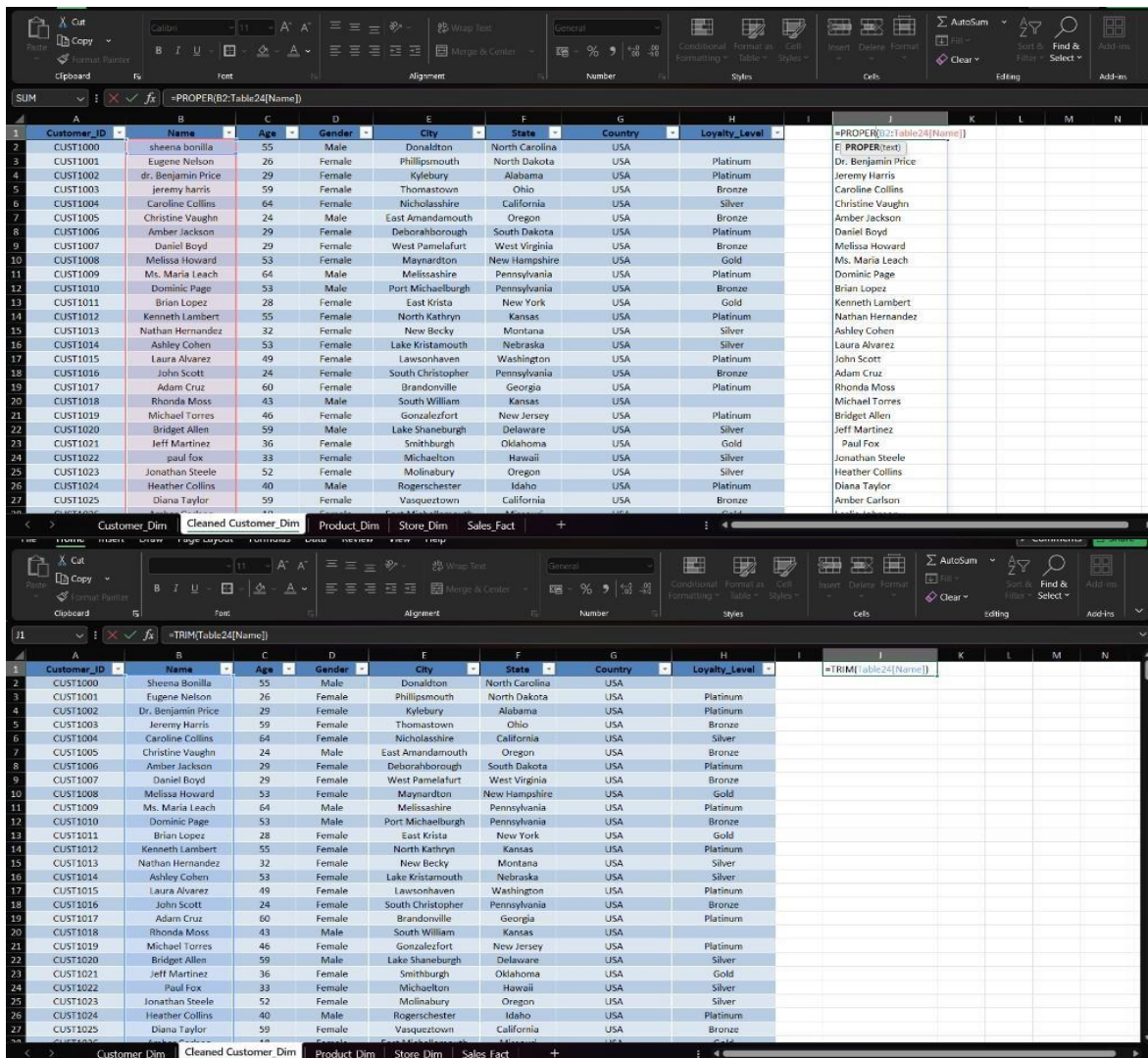
DATA CLEANING

A) In Customer_Dim table, the inconsistent datas of CUSTOMER_ID column format is changed Using the FIND AND REPLACE Option.

Eg: C-1464---- CUST1464



B) The NAME column of the Customer_Dim table is changed using the TRIM () AND PROPER () function.



c) The Missing values of the LOYALTY_LEVEL column of the Customer_Dim table are filled with PLATINUM by using the COUNTIF () and MODE () function.

Customer_ID	Name	Age	Gender	City	State	Country	Loyalty_Level
CUST1000	Sheena Bonilla	55	Male	Donaldfon	North Carolina	USA	Platinum
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA	Platinum
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA	Platinum
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA	Gold
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	52	Female	Mollinabury	Oregon	USA	Silver
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	59	Female	Vasqueztown	California	USA	Bronze
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA	Gold
CUST1027	Leslie Johnson	31	Male	Mollyshire	Utah	USA	Platinum
CUST1028	Joshua Ryan	27	Female	New David	New Jersey	USA	Platinum
CUST1029	David Arnold	54	Female	East Jennifer	California	USA	Silver
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CUST1031	Sarah Nieves	41	Male	Robertville	Idaho	USA	Silver

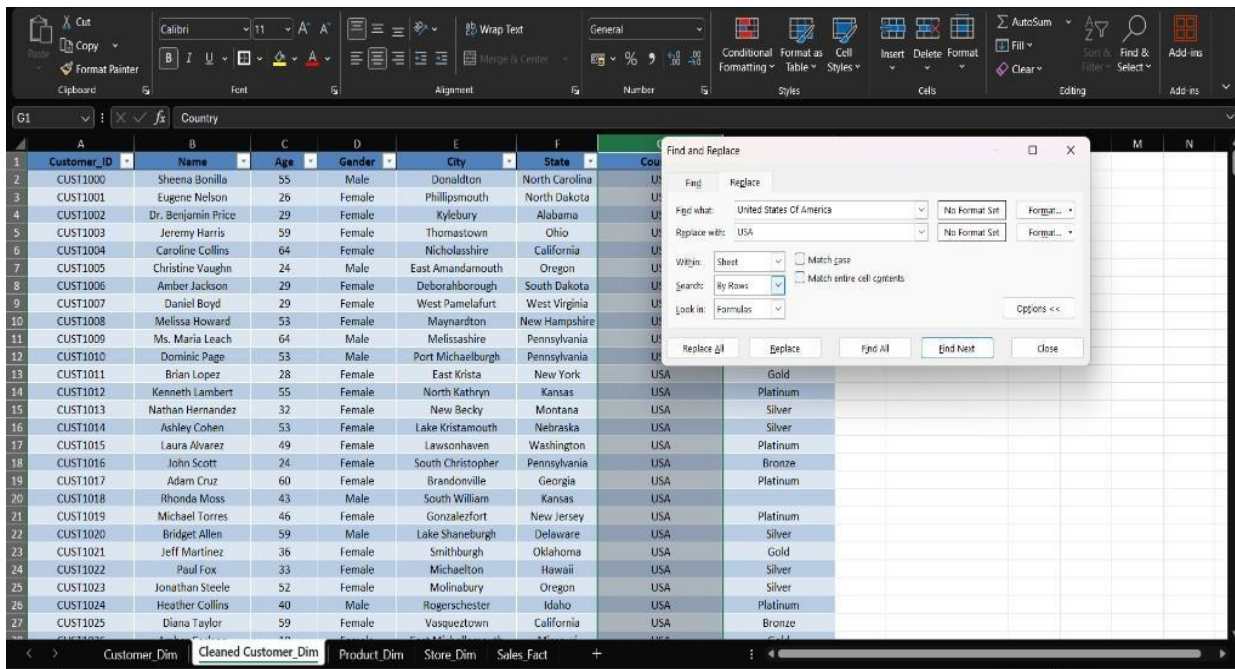
Loyalty_Level	Count
Platinum	149
Gold	122
Silver	104

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CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA	Platinum
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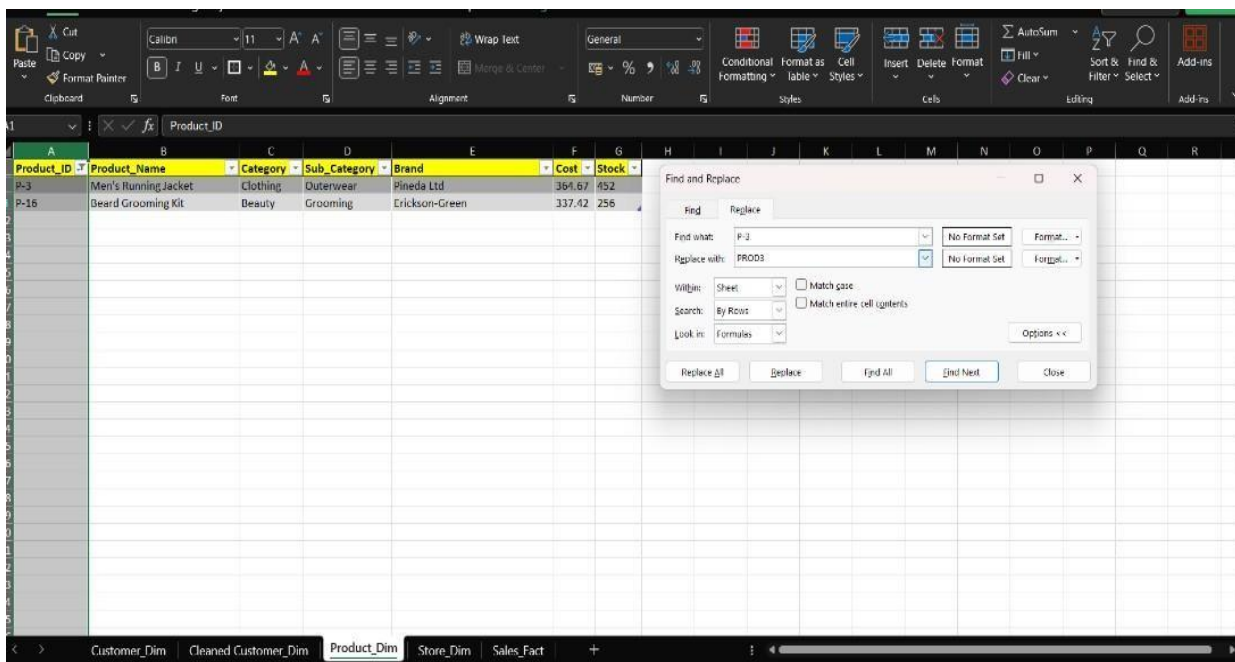
Here , the count of PLATINUM is higher , so the missing values are filled with PLATINUM.

D) The Inconsistency of COUNTRY column in the Customer_dim table is corrected using the FIND AND REPLACE Option.

Eg: United States Of America ---- USA , usa----USA



E) In the Product_Dim table, the inconsistencies of PRODUCT_ID column is corrected using the FIND AND REPLACE option.
Eg: P-3---PROD3 , P-16---PROD16



F) The missing values of the COST AND STOCK columns in the Product_Dim table is Corrected using the AVERAGEIFS () function with the CATEGORY and SUB_CATEGORY columns.

SUM										=AVERAGEIFS(Table467[Cost],Table467[Category],C4,Table467[Sub_Category],D4)									
Product_ID		Product_Name		Category	Sub_Category	Brand	Cost	Stock											
PROD00	Brother Laser Printer		Electronics	Computer	Hunt Inc	112.42	416												
PROD01	Men's Graphic T-Shirt		Clothing	Tops	Terry-Moore	342.36	181	=AVERAGEIFS(Table467[Cost],Table467[Category],C4,Table467[Sub_Category],D4)											
PROD02	Women's Casual Hoodie		Clothing	Outerwear	Sims Inc	296.82	352												
PROD04	Men's Running Jacket		Clothing	Outerwear	Pineda Ltd	364.67		AVERAGEIFS(average_range, criteria_range1, [criteria_range2, criteria2], [criteria_range3, criteria3])											
PROD04	Women's Yoga Pants		Clothing	Bottoms	Gonzalez-Davis	306.46	487												
PROD05	Men's Leather Belt		Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59												
PROD06	Herbal Face Moisturizer		Beauty	Skincare	Davidson PLC	312.8	350												
PROD07	LED Table Lamp		Home	Lighting	Harrell LLC	218.86	170												
PROD08	Wireless Bluetooth Earbuds		Electronics	Audio	Perez-Johnston	439.07	142												
PROD09	Adjustable Dumbbell Set		Sports	Fitness	Hill PLC	495.72	323												
PROD10	Ceramic Kitchen Bowl Set		Home	Kitchen	Hamilton Group	298.86	308												
PROD11	HD Smart TV Stick		Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	338												
PROD12	Hair Strengthening Shampoo		Beauty	Haircare	Rice LLC	115.82	460												
PROD13	USB-C Fast Charger		Electronics	Accessories	Dawson-Perkins	242.23	453												
PROD14	Decorative Wall Mirror		Home	Decor	Luna, Anderson and Bowman	48.01	439												
PROD15	4K Home Projector		Electronics	Appliances	Thomas LLC	465.34	301												
PROD16	Board Grooming Kit		Beauty	Grooming	Frickson-Green	337.42	256												
PROD17	Soccer Training Cone Set		Sports	Accessories	Ford, Murphy and Wilson	144.53	367												
PROD18	Women's Denim Jeans		Clothing	Bottoms	Guerra-Hendrix	303.75	343												
PROD19	Knee Support Brace		Sports	Equipment	May-Reese	21.34	65												
PROD20	Outdoor Sports Water Bottle		Sports	Accessories	Soto-Ellis	131.51	199.2												
PROD21	Multi-Purpose Storage Box		Home	Storage	Gardner-Montes	123.46	76												
PROD22	Carbon Fiber Tennis Racket		Sports	Equipment	Flores, Pineda and Rodriguez	484.62	432												
PROD23	Men's Formal Shirt		Clothing	Tops	Moon, Mora and Santos	283	360												
PROD24	Foldable Camping Chair		Sports	Outdoor	Cook Ltd	166.06	340												
PROD25	Women's Scarf Set		Clothing	Accessories	Brown and Sons	174.02	268												

Cut Copy Paste		B I U		Align Center		Number		Conditional Formatting		Cell Styles		Insert Delete Format		AutoSum Fill Clear		Find & Select		Add-ins	
Clipboard		Font		Alignment		Number		Styles		Cells				Editing					
SUM																			
=AVERAGEIFS(Table467[Stock],Table467[Category],C13,Table467[Sub_Category],D13)																			
A	B	C	D	E	F	G	H	I	J	K	L	M	N	O					
Product_ID	Product_Name	Category	Sub_Category	Brand	Cost	Stock													
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PROD05	Women's Yoga Pants	Clothing	Bottoms	Gonzalez-Davis	306.46	487													
PROD06	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	59													
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Product_Dim Sheet2 Store_Dim Sales_Fact Cleared_Customer_Dim Cleared_Product_Dim Clean																			

G) The Store_Dim has no inconsistent datas and are kept as it is for the Future use.

H) In the Sales_fact table , the missing values of QUANTITY is calculated using the Formula

$$\text{QUANTITY} = \text{TOTAL AMOUNT} / (\text{UNIT_PRICE} * (1 - \text{DISCOUNT}))$$

ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
2	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	0.09	Credit Card	198.4164
3	2023-12-25	CUST1467	PROD74	STORE17	3	104.47	0.27	PayPal	278.7893
4	2024-06-03	CUST1056	PROD49	STORE8	4	914.05	0.25	Credit Card	2742.15
5	2023-11-29	CUST1346	PROD76	STORE8	1	921.51	0.16	COD	774.0684
6	2024-01-06	CUST1493	PROD59	STORE5	3	745.3	0.18	Credit Card	1833.438
7	2024-07-26	CUST1116	PROD15	STORE7	3	147.01	0.19	COD	357.2343
8	2025-07-12	CUST1016	PROD56	STORE4	2	785.31	0.15	PayPal	1335.027
9	2023-11-18	CUST1152	PROD70	STORE8	4	342.39	0.23	PayPal	1054.5612
10	2025-05-04	CUST1387	PROD99	STORE5	1	648.25	0.07	PayPal	1250.6175
11	2025-07-31	CUST1464	PROD33	STORE0	2	873.17	0.23	PayPal	1344.6818
12	2024-02-13	CUST1065	PROD24	STORE14	1	278.63	0.07	Credit Card	259.1259
13	2025-05-24	CUST1320	PROD39	STORE1	4	286.92	0.17	Credit Card	952.5744
14	2024-08-24	CUST1237	PROD88	STORE4	4	644.58	0.01	Credit Card	2552.5368
15	2025-08-06	CUST1443	PROD14	STORE16	2	316.9	0.3	PayPal	443.86
16	2025-03-21	CUST1348	PROD24	STORE15	4	403.47	0.29	PayPal	1145.8548
17	2024-12-10	CUST1435	PROD36	STORE9	1	871.34	0.04	COD	2509.4592
18	2025-09-13	CUST1276	PROD65	STORE19	2	451.98	0.25	Credit Card	677.97
19	2024-12-11	CUST1079	PROD27	STORE4	2	502.69	0.23	Credit Card	774.1426
20	2024-08-13	CUST1094	PROD39	STORE4	2	559.23	0.25	COD	838.845
21	2024-03-18	CUST1239	PROD60	STORE13	4	1782.4512	0.04	COD	1782.4512
22	2025-07-05	CUST1144	PROD72	STORE17	3	986.01	0.27	COD	2159.4276
23	2024-06-04	CUST1396	PROD43	STORE18	1	589.56	0.22	Credit Card	459.8568
24	2024-04-12	CUST1476	PROD69	STORE10	3	979.42	0.03	PayPal	2850.1122
25	2024-06-23	CUST1368	PROD12	STORE8	1	719.28	0.08	Credit Card	661.7376
26	2025-09-08	CUST1186	PROD67	STORE6	3	77.6	0.07	Credit Card	216.504
27	2024-09-17	CUST1381	PROD5	STORE3	4	852.71	0.14	PayPal	2933.3224
28	2024-01-03	CUST1343	PROD95	STORE9	4	456.46	0.07	Credit Card	500.4144

I) In the Sales_fact table , the missing values of UNIT PRICE is calculated using the Formula

$$\text{UNIT_PRICE} = \text{TOTAL AMOUNT} / (\text{QUANTITY} * (1 - \text{DISCOUNT}))$$

ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
21	2024-03-18	CUST1239	PROD60	STORE13	4		0.04	COD	1782.4512
54	2025-08-25	CUST1148	PROD89	STORE6	3			Credit Card	486.0633
270	2024-09-20	CUST1406	PROD90	STORE18	3		0.06	Credit Card	315.2478

J) In the Sales_fact table , the missing values of DISCOUNT is calculated using the Formula

$$\text{DISCOUNT} = 1 - (\text{TOTAL AMOUNT} / (\text{QUANTITY} * \text{UNIT_PRICE}))$$

	A	B	C	D	E	F	G	H	I	J	K	L	M
	Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount			
28	SALE26	2024-01-03	CUST1213	PROD98	STORE8	4	156.16	0.04	Credit Card	599.6544			
54	SALE52	2025-08-25	CUST1148	PROD89	STORE6	3			Credit Card	486.0633			
60	SALE58	2024-12-07	CUST1404	PROD15	STORE7	4	733.42	0.13	PayPal	2552.3016			
261	SALE259	2025-04-26	CUST1248	PROD35	STORE2	2	355.3	0.17	PayPal	589.798			

To calculate the DISCOUNT

=1-(F261/F261*G261)

K) The Sales_fact table , the missing values of TOTAL_AMOUNT is calculated using the Formula

$$\text{TOTAL_AMOUNT} = (\text{QUANTITY} * \text{UNIT_PRICE}) * (1 - \text{DISCOUNT})$$

	A	B	C	D	E	F	G	H	I	J	K	L	M
	Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount			
66	SALE64	2025-09-14	CUST1255	PROD53	STORE10	3	480.06	0.01	Credit Card				
72	SALE70	2025-07-11	CUST1058	PROD60	STORE5	3	92.24	0.28	COD				
86	SALE84	2023-12-08	CUST1423	PROD46	STORE17	1	773.57	0.05	COD				
96	SALE94	2024-08-26	CUST1298	PROD97	STORE19	2	569.36	0.01	PayPal				

to calculate the TOTAL AMOUNT

=F66*G66*(1-H66)

L) In the 54th row of Sales_Fact Table, the values of UNIT_PRICE and DISCOUNT IS MISSING. So the DISCOUNT value is corrected using the AVERAGE () function and then the UNIT_PRICE is calculated using the formula as mentioned above.

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	218.04	9%	Credit Card	198.4164
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	104.47	27%	PayPal	228.7893
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	914.05	25%	Credit Card	2742.15
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	921.51	16%	COD	774.0684
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	745.3	18%	Credit Card	1833.438
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	147.01	19%	COD	357.2343
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	785.31	15%	PayPal	1335.027
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	342.39	23%	PayPal	1054.5612
SALE8	04-05-2025	CUST1187	PROD99	STORE5	3	448.25	7%	PayPal	1250.6175
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	873.17	23%	PayPal	1344.6818
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	278.63	7%	Credit Card	259.1259
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	286.92	17%	Credit Card	952.5744
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	644.58	1%	Credit Card	2552.5368
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	316.9	30%	PayPal	443.66
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	403.47	29%	PayPal	1145.8548
SALE15	10-12-2024	CUST1435	PROD36	STORE9	3	871.34	4%	COD	2509.4592
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	451.98	25%	Credit Card	677.97
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	502.69	23%	Credit Card	774.1426
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	559.23	25%	COD	838.845
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	464.18	4%	COD	1782.4512
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	986.04	27%	COD	2159.4276
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	589.56	22%	Credit Card	459.8568
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	979.42	3%	PayPal	2850.1122
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	719.28	8%	Credit Card	661.7376
SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	77.6	7%	Credit Card	216.504
SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	852.71	14%	PayPal	2933.3224

M) The Data_type of the ORDER_DATE and DISCOUNT is Formatted into Short Date and Percentage.

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	9%	Credit Card	198.4164
SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	104.47	27%	PayPal	228.7893
SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	914.05	25%	Credit Card	2742.15
SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	921.51	16%	COD	774.0684
SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	745.3	18%	Credit Card	1833.438
SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	147.01	19%	COD	357.2343
SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	785.31	15%	PayPal	1335.027
SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	342.39	23%	PayPal	1054.5612
SALE8	2025-05-04	CUST1187	PROD99	STORE5	3	448.25	7%	PayPal	1250.6175
SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	873.17	23%	PayPal	1344.6818
SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	278.63	7%	Credit Card	259.1259
SALE11	2025-05-24	CUST1320	PROD39	STORE1	4	286.92	17%	Credit Card	952.5744
SALE12	2024-08-24	CUST1237	PROD88	STORE4	4	644.58	1%	Credit Card	2552.5368
SALE13	2025-08-06	CUST1443	PROD34	STORE16	2	316.9	30%	PayPal	443.66
SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	403.47	29%	PayPal	1145.8548
SALE15	2024-12-10	CUST1435	PROD36	STORE9	3	871.34	4%	COD	2509.4592
SALE16	2025-09-13	CUST1276	PROD65	STORE19	2	451.98	25%	Credit Card	677.97
SALE17	2024-12-11	CUST1079	PROD27	STORE4	2	502.69	23%	Credit Card	774.1426
SALE18	2024-08-13	CUST1094	PROD39	STORE4	2	559.23	25%	COD	838.845
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4	464.18	4%	COD	1782.4512
SALE20	2025-07-05	CUST1144	PROD72	STORE17	3	986.04	27%	COD	2159.4276
SALE21	2024-06-04	CUST1396	PROD43	STORE18	1	589.56	22%	Credit Card	459.8568
SALE22	2024-04-12	CUST1476	PROD69	STORE10	3	979.42	3%	PayPal	2850.1122
SALE23	2024-06-23	CUST1368	PROD12	STORE8	1	719.28	8%	Credit Card	661.7376
SALE24	2025-09-08	CUST1186	PROD67	STORE6	3	77.6	7%	Credit Card	216.504
SALE25	2024-09-17	CUST1381	PROD5	STORE3	4	852.71	14%	PayPal	2933.3224

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount
SALE0	07-07-2024	CUST1409	PROD97	STORE3	1	218.04	9%	Credit Card	198.4164
SALE1	25-12-2023	CUST1467	PROD74	STORE17	3	104.47	27%	PayPal	228.7893
SALE2	03-06-2024	CUST1056	PROD49	STORE8	4	914.05	25%	Credit Card	2742.15
SALE3	29-11-2023	CUST1346	PROD76	STORE8	1	921.51	16%	COD	774.0684
SALE4	06-01-2024	CUST1493	PROD59	STORE5	3	745.3	18%	Credit Card	1833.438
SALE5	26-07-2024	CUST1116	PROD15	STORE7	3	147.01	19%	COD	357.2343
SALE6	12-07-2025	CUST1016	PROD56	STORE4	2	785.31	15%	PayPal	1335.027
SALE7	18-11-2023	CUST1152	PROD70	STORE8	4	342.39	23%	PayPal	1054.5612
SALE8	04-05-2025	CUST1187	PROD99	STORE5	3	448.25	7%	PayPal	1250.6175
SALE9	31-07-2025	CUST1464	PROD33	STORE0	2	873.17	23%	PayPal	1344.6818
SALE10	13-02-2024	CUST1065	PROD24	STORE14	1	278.63	7%	Credit Card	259.1259
SALE11	24-05-2025	CUST1320	PROD39	STORE1	4	286.92	17%	Credit Card	952.5744
SALE12	24-08-2024	CUST1237	PROD88	STORE4	4	644.58	1%	Credit Card	2552.5368
SALE13	06-08-2025	CUST1443	PROD34	STORE16	2	316.9	30%	PayPal	443.66
SALE14	21-06-2025	CUST1348	PROD24	STORE15	4	403.47	29%	PayPal	1145.8548
SALE15	10-12-2024	CUST1435	PROD36	STORE9	3	871.34	4%	COD	2509.4592
SALE16	13-09-2025	CUST1276	PROD65	STORE19	2	451.98	25%	Credit Card	677.97
SALE17	11-12-2024	CUST1079	PROD27	STORE4	2	502.69	23%	Credit Card	774.1426
SALE18	13-08-2024	CUST1094	PROD39	STORE4	2	559.23	25%	COD	838.845
SALE19	18-03-2024	CUST1239	PROD60	STORE13	4	464.18	4%	COD	1782.4512
SALE20	05-07-2025	CUST1144	PROD72	STORE17	3	986.04	27%	COD	2159.4276
SALE21	04-06-2024	CUST1396	PROD43	STORE18	1	589.56	22%	Credit Card	459.8568
SALE22	12-04-2024	CUST1476	PROD69	STORE10	3	979.42	3%	PayPal	2850.1122
SALE23	23-06-2024	CUST1368	PROD12	STORE8	1	719.28	8%	Credit Card	661.7376
SALE24	08-09-2025	CUST1186	PROD67	STORE6	3	77.6	7%	Credit Card	216.504
SALE25	17-09-2024	CUST1381	PROD5	STORE3	4	852.71	14%	PayPal	2933.3224

STEP-3

After the cleaning of Datas, The Tables are renamed as Cleaned Customer_Dim, Cleaned Product_Dim, Cleaned Store_Dim, Cleaned Sales_Fact.

Now , We can create the Combined table Using VLOOKUP () and XLOOKUP () function and are naming it as VLOOKUP() & XLOOKUP().

The contents of the Cleaned Sales_Fact table is copied into the VLOOKUP() & XLOOKUP() table , and the datas in the Cleaned Customer_Dim and Cleaned Product_Dim are combined using VLOOKUP () function and the datas in the Cleaned Store_Dim table is combined using XLOOKUP () function.

Sales_ID	Order_Date	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Customer_Name
SALE0	2024-07-07	CUST1409	PROD97	STORE3	1	218.04	9%	Credit Card	198.4164	Sarah Smith
SALE1	2023-12-25	CUST1467	PROD74	STORE17	3	104.47	27%	PayPal	228.7893	Michael Foley
SALE2	2024-06-03	CUST1056	PROD49	STORE8	4	914.05	25%	Credit Card	2742.15	Christopher Sutton
SALE3	2023-11-29	CUST1346	PROD76	STORE8	1	921.51	16%	COD	774.0684	John Jordan
SALE4	2024-01-06	CUST1493	PROD59	STORE5	3	745.3	18%	Credit Card	1833.438	Amanda Powell
SALE5	2024-07-26	CUST1116	PROD15	STORE7	3	147.01	19%	COD	357.2343	Anthony Phillios
SALE6	2025-07-12	CUST1016	PROD56	STORE4	2	785.31	15%	PayPal	1335.027	John Scott
SALE7	2023-11-18	CUST1152	PROD70	STORE8	4	342.39	23%	PayPal	1054.5612	Patrick Roberts
SALE8	2025-05-04	CUST1187	PROD99	STORE5	3	448.25	7%	PayPal	1250.6175	Dr. Suzanne Bullock
SALE9	2025-07-31	CUST1464	PROD33	STORE0	2	873.17	23%	PayPal	1344.6818	David Pierce
SALE10	2024-02-13	CUST1065	PROD24	STORE14	1	278.63	7%	Credit Card	259.1259	Michael Thomas
SALE11	2025-05-24	CUST1320	PROD39	STORE1	4	286.92	17%	Credit Card	952.5744	Jessica Gonzalez
SALE12	2024-08-24	CUST1237	PROD88	STORE4	4	644.58	1%	Credit Card	2552.5368	Wesley Short
SALE13	2025-08-06	CUST1443	PROD34	STORE16	2	316.9	30%	PayPal	443.66	Adam Castro
SALE14	2025-06-21	CUST1348	PROD24	STORE15	4	403.47	29%	PayPal	1145.8548	Sherry Cabrera
SALE15	2024-12-10	CUST1435	PROD36	STORE9	3	871.34	4%	COD	2509.4592	Lindsay Beasley
SALE16	2025-09-13	CUST1276	PROD65	STORE19	2	451.98	25%	Credit Card	677.97	Arthur Warren
SALE17	2024-12-11	CUST1079	PROD27	STORE4	2	502.69	23%	Credit Card	774.1426	Molly Gardner
SALE18	2024-08-13	CUST1094	PROD39	STORE4	2	559.23	25%	COD	838.845	David Patterson
SALE19	2024-03-18	CUST1239	PROD60	STORE13	4	464.18	4%	COD	1782.4512	Mario Hill
SALE20	2025-07-05	CUST1144	PROD72	STORE17	3	986.04	27%	COD	2159.4276	Justin Cross
SALE21	2024-06-04	CUST1396	PROD43	STORE18	1	589.56	22%	Credit Card	459.8568	Alexander Byrd
SALE22	2024-04-12	CUST1476	PROD69	STORE10	3	979.42	3%	PayPal	2850.1122	Elizabeth Foster
SALE23	2024-06-23	CUST1368	PROD12	STORE8	1	719.28	8%	Credit Card	661.7376	Miranda King
SALE24	2025-09-08	CUST1186	PROD67	STORE6	3	77.6	7%	Credit Card	216.504	Jennifer Pruitt
SALE25	2024-09-17	CUST1381	PROD05	STORE3	4	852.71	14%	PayPal	2933.3224	Erika Padilla

The same procedure is followed for the AGE,GENDER,STATE,LOYALTY_LEVEL columns of the Cleaned Customer_Dim table.

H	I	J	K	L	M	N	O	P	Q	R	S	T
1	entity	Unit_Price	Discount	Payment_Type	Total_Amount	Customer_Name	Age	Gender	Customer_City	State	Loyalty_Level	Product_Name
2	1	218.04	9%	Credit Card	198.4164	Sarah Smith	55	Male	Kathleenport	Wyoming	Gold	Cricket Bat
3	3	104.47	27%	PayPal	228.7893	Michael Foley	63	Female	Oneillfurt	Vermont	Gold	Bluetooth Car Audio System
4	4	914.05	25%	Credit Card	2742.15	Christopher Sutton	49	Female	Deborahberg	Pennsylvania	Gold	Men's Roll-On Perfume
5	1	921.51	16%	COD	774.0684	John Jordan	58	Female	Port Crystalberg	Tennessee	Platinum	Wireless Mouse
6	3	745.3	18%	Credit Card	1833.438	Amanda Pouell	28	Male	South William	Alabama	Silver	Men's Hair Styling Gel
7	3	147.01	19%	COD	357.2343	Anthony Phillips	18	Female	North Davidport	Rhode Island	Platinum	4K Home Projector
8	2	785.31	15%	PayPal	1335.027	John Scott	24	Female	South Christopher	Pennsylvania	Bronze	Premium Blanket Throw
9	4	342.39	23%	PayPal	1054.5612	Patrick Roberts	41	Female	Jaredville	Missouri	Platinum	Kitchen Storage Rack
10	3	448.25	7%	PayPal	1250.6175	Dr. Suzanne Bullock	60	Female	Lake Blaketon	Alabama	Platinum	Stainless Steel Cookware Set
11	2	873.17	23%	PayPal	1344.6818	David Pierce	45	Female	Grayland	Washington	Platinum	Modern Wall Clock
12	1	278.63	7%	Credit Card	259.1259	Michael Thomas	21	Female	Woodmouth	Nevada	Silver	Foldable Camping Chair
13	4	286.92	17%	Credit Card	952.5744	Jessica Gonzalez	34	Female	South Darrenland	Iowa	Bronze	Portable Power Bank
14	4	644.58	1%	Credit Card	2592.5368	Wesley Short	41	Female	Susanstad	Montana	Gold	Sports Wrist Sweatbands
15	2	316.9	30%	PayPal	443.66	Adam Castro	51	Male	New Donnanbury	Texas	Silver	Gaming Mechanical Keyboard
16	4	403.47	29%	PayPal	1145.8548	Sherry Cabrera	55	Male	Shannonview	Pennsylvania	Silver	Foldable Camping Chair
17	3	871.34	4%	COD	2509.4592	Lindsay Beasley	37	Female	Wesleyhaven	Idaho	Silver	Women's Summer Dress
18	2	451.98	25%	Credit Card	677.97	Arthur Warren	38	Male	Erichshire	Nevada	Gold	Adjustable Dumbbells 20kg
19	2	502.69	23%	Credit Card	774.1426	Molly Gardner	22	Female	Nicolborough	Arkansas	Platinum	Decorative Floor Vase
20	2	559.23	25%	COD	838.845	David Patterson	43	Female	New Amyport	Nevada	Gold	Portable Power Bank
21	4	464.18	4%	COD	1782.4512	Maria Hill	58	Female	Zamoraborough	Nevada	Platinum	Luxury Perfume Spray
22	3	986.04	27%	COD	2159.4276	Justin Cross	62	Female	North Anthonychester	Nevada	Bronze	Professional Jump Rope
23	1	589.56	22%	Credit Card	459.8568	Alexander Byrd	25	Female	Parkerview	New York	Silver	USB External Hard Drive
24	3	979.42	3%	PayPal	2850.1122	Elizabeth Byrd	33	Female	Anthonymouth	Oklahoma	Gold	Men's Sports Shorts
25	1	719.28	8%	Credit Card	661.7376	Miranda King	34	Female	West Mathew	New York	Platinum	Hair Strengthening Shampoo
26	3	77.6	7%	Credit Card	216.504	Jennifer Pruitt	53	Male	East Justin	Missouri	Silver	Smart LED Light Strip
27	4	852.71	14%	PayPal	2933.3274	Erika Padilla	50	Male	Andrewsland	Alabama	Silver	Men's Leather Belt
28	4	156.16	400%	Credit Card	599.6544	Stuart Johnson	42	Female	West Markport	North Dakota	Platinum	Portable Bluetooth Speaker
29	3	815.49	17%	Credit Card	2030.5701	Anna Perez	30	Male	South Jerry	Maine	Platinum	WiFi Mesh Router
30	2	74.21	11%	PayPal	132.0938	Theresa Campbell	42	Female	New Brian	Wisconsin	Bronze	USB-C Fast Charger
31	4	451.67	15%	Credit Card	1535.678	Sherry Cabrera	55	Male	Shannonview	Pennsylvania	Silver	Stainless Steel Cookware Set
32	1	429.38	13%	PayPal	373.5606	Joshua Ryan	27	Female	New David	New Jersey	Platinum	Men's Leather Belt
33	2	148.06	1%	COD	788.9824	Khana Rinnam	73	Female	East Andrus	Michigan	Gold	Men's Track Pants

The same procedure is followed for
CATEGORY, SUB_CATEGORY, BRAND, COST and STOCK columns of the
Cleaned Product_Dim table.

N	O	P	Q	R	S	T	U	V	W	X	Y
1	Age	Gender	State	Loyalty_Level	Product_Name	Category	Sub_Category	Brand	cost	Store_Name	Region
2	55	Male	Wyoming	Gold	Cricket Bat	Sports	Team Sports	Khan LLC	245.98	Johnson-Wright	North
3	63	Female	Vermont	Gold	Bluetooth Car Audio System	Electronics	Audio	Bruce Bowen	403.71	Holloway Ltd	North
4	49	Female	Pennsylvania	Gold	Men's Roll-On Perfume	Beauty	Fragrance	Sullivan and Sons	43.79	Bean, Jones and Park	West
5	58	Female	Tennessee	Platinum	Wireless Mouse	Electronics	Computer	Larson Ltd	392.89	Bean, Jones and Park	West
6	28	Male	Alabama	Silver	Men's Hair Styling Gel	Beauty	Haircare	Cline McCarthy	17.31	Gray Ltd	North
7	18	Female	Rhode Island	Platinum	4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	Martinez-Mullins	South
8	24	Female	Pennsylvania	Bronze	Premium Blanket Throw	Home	Decor	Hernandez, Kennedy and Porter	461.47	Alexander Group	South
9	41	Female	Missouri	Platinum	Kitchen Storage Rack	Home	Storage	Duran Group	138.05	Bean, Jones and Park	West
10	60	Female	Alabama	Platinum	Stainless Steel Cookware Set	Home	Kitchen	Campbell Ltd	201.41	Gray Ltd	North
11	45	Female	Washington	Platinum	Modern Wall Clock	Home	Decor	Weiss-Watson	179.77	Rivera-Warner	East
12	21	Female	Nevada	Silver	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	McDonald Inc	South
13	34	Female	Iowa	Bronze	Portable Power Bank	Electronics	Mobile	Cummings and Sons	326.06	Martinez, Adkins and Johnson	North
14	41	Female	Montana	Gold	Sports Wrist Sweatbands	Sports	Accessories	Olsen-Daniels	131.24	Alexander Group	South
15	51	Male	Texas	Silver	Gaming Mechanical Keyboard	Electronics	Computer	Lucero LLC	419.28	Frederick-Wright	West
16	55	Male	Pennsylvania	Silver	Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	Shelton PLC	North
17	37	Female	Idaho	Silver	Women's Summer Dress	Clothing	Tops	Smith-Mills	223.43	Moore, Padilla and Butler	South
18	38	Male	Nevada	Gold	Adjustable Dumbbells, 20kg	Sports	Fitness	Montgomery, Benton and Miller	240.63	Mann-Austin	East
19	22	Female	Arkansas	Platinum	Decorative Floor Vase	Home	Decor	Khan, Jones and Paul	219.12	Alexander Group	South
20	43	Female	Nevada	Gold	Portable Power Bank	Electronics	Mobile	Cummings and Sons	326.06	Alexander Group	South
21	58	Female	Nevada	Platinum	Luxury Perfume Spray	Beauty	Fragrance	Gibson, Green and Lopez	487.75	Gardner, Humphrey and Contreras	East
22	62	Female	Nevada	Bronze	Professional Jump Rope	Sports	Fitness	Davis PLC	477.61	Holloway Ltd	North
23	25	Female	New York	Silver	USB External Hard Drive	Electronics	Computer	Robinson Inc	253.33	Guerreiro Schmitt	East
24	33	Female	Oklahoma	Gold	Men's Sports Shorts	Clothing	Bottoms	Fitzpatrick Ltd	218.72	Hill, Wagner and Ballard	East
25	34	Female	New York	Platinum	Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	Bean, Jones and Park	West
26	53	Male	Missouri	Silver	Smart LED Light Strip	Electronics	Accessories	Flores Davis	349.17	Benson-Elliott	East
27	50	Male	Alabama	Silver	Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	Johnson-Wright	North

The same Procedure is followed for the
REGION, CITY and STORE_TYPE columns of the **Cleaned Store_Dim** table.
 The New columns are created .
 For Eg:
GROSS_SALES,DISCOUNT_AMOUNT,PROFIT,TOTAL_COST,
PROFIT_MARGIN ,AGE_GROUP using the formulas.

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design								
AH2 =IF([@[Total_Amount]]=0,0,[@Profit]/[@[Total_Amount]])								
	AA	AB	AC	AD	AE	AF	AG	AH
1	City	Store_Type	Gross_Sales	Discount_Amount	Total_Cost	Profit	Age_Group	Profit_Margin
2	Hoodberg	Outlet	218.04	19.6236	245.98	-47.5636	46-55	-0.239716072
3	Bensonhaven	Flagship	313.41	84.6207	1211.13	-982.3407	56-64	-4.293647911
4	Thompsonborough	Online	3656.2	914.05	173.16	2568.99	46-55	0.93685247
5	Thompsonborough	Online	921.51	147.4416	392.89	381.1784	56-64	0.492435035
6	New Michaelchester	Outlet	2235.9	402.462	51.93	1781.508	26-35	0.971676162
7	Williamsfurt	Online	441.03	83.7957	1396.02	-1038.7857	18-25	-2.907855433
8	Port Sharon	Online	1570.62	235.593	922.94	412.087	18-25	0.308673158
9	Thompsonborough	Online	1369.56	314.9988	552.2	502.3612	36-45	0.476369887
10	New Michaelchester	Outlet	1344.75	94.1325	604.23	646.3875	56-64	0.516854674
11	Washingtonfurt	Flagship	1746.34	401.6582	359.54	985.1418	36-45	0.732620758
12	Christopherland	Outlet	278.63	19.5041	166.06	93.0659	18-25	0.359153215
13	Danielmouth	Flagship	1147.68	195.1056	1304.24	-351.6656	26-35	-0.369173893
14	Port Sharon	Online	2578.32	25.7832	524.96	2027.5768	36-45	0.794337931
15	New Jimmymouth	Flagship	633.8	190.14	838.56	-394.9	46-55	-0.890096019
16	East Jenniferstad	Flagship	1613.88	468.0252	664.24	481.6148	46-55	0.420310497
17	East Johnstad	Flagship	2614.02	104.5608	670.29	1839.1692	36-45	0.732894641
18	Vasquezborough	Online	903.96	225.99	481.26	196.71	36-45	0.290145582
19	Port Sharon	Online	1005.38	231.2374	438.24	335.9026	18-25	0.433902746
20	Port Sharon	Online	1118.46	279.615	652.12	186.725	36-45	0.222597739
21	Johnsonville	Flagship	1856.72	74.2688	1931	-148.5488	56-64	-0.083339617
22	Bensonhaven	Flagship	2958.12	798.6924	1432.83	726.5976	56-64	0.336476944
23	East Stevechester	Flagship	589.56	129.7032	253.33	206.5268	18-25	0.449111115
24	East Davidville	Online	2938.26	88.1478	654.66	2195.4522	26-35	0.77030378
25	Thompsonborough	Online	719.28	57.5424	115.82	545.9176	26-35	0.824975942
26	Port Josephberg	Online	232.8	16.296	1047.51	-831.006	46-55	-3.838293981
27	Hoodberg	Outlet	3410.84	477.5176	1044	1889.3224	46-55	0.644089583
28	Thompsonborough	Online	624.64	2498.56	1726.68	-1127.0256	36-45	-1.879458568
29	Thompsonborough	Online	2446.47	415.8999	1442.7	587.8701	26-35	0.289509877
30	North Nancyview	Outlet	148.42	16.3262	484.46	-352.3662	36-45	-2.667545335
31	Hoodberg	Outlet	1806.68	271.002	805.64	730.038	46-55	0.475384814
32	Vasquezborough	Online	429.38	55.8194	261	112.5606	26-35	0.30131818
33	Schrodtown	Online	907.02	8.9376	773.12	-484.1376	18-25	-1.675318635

STEP-4

Descriptive Statistics using Analysis Toolpak is performed for each table.

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design

Get Data From Text/CSV Recent Sources From Web Existing Connections Refresh All Workbook Links Queries & Connections Properties Sort & Filter Filter Clear Apply Text to Columns Data Validation Remove Duplicates Data Model Consolidate What-If Analysis Forecast Subtotal Outline Data Analysis

Customer_ID

Customer_ID	Name	Age	Gender	City	State	Country
CUST1476	Elizabeth Foster	33	Female	Anthonytown	Oklahoma	USA
CUST1477	Andrea Harris	41	Female	West Kelseyberg	Kansas	USA
CUST1478	Earl Smith Dds	59	Male	West Sarah	South Dakota	USA
CUST1479	Dylan Gutierrez	40	Male	New Jamesview	West Virginia	USA
CUST1480	Jordan Medina	33	Male	Robinchester	Florida	USA
CUST1481	Bryan Tran	21	Male	Kelseybury	Georgia	USA
CUST1482	Sheri Joyce	26	Male	West Tylerfort	Kentucky	USA
CUST1483	Sarah Smith	48	Male	East Rodney	Nevada	USA
CUST1484	Jessica Castro	32	Male	West Markton	Massachusetts	USA
CUST1485	Sara Ingram	46	Female	East Dillonview	Virginia	USA
CUST1486	Eric Lee	22	Female	Lake April	Vermont	USA
CUST1487	Amanda Medina	36	Female	East Patriciaad	Utah	USA
CUST1488	Zachary Andrade	23	Female	South Davidport	California	USA
CUST1489	Melinda Thomas	42	Female	Lake Elizabethside	Indiana	USA
CUST1490	Scott Harris	23	Male	North Nataliemouth	Maryland	USA
CUST1491	Kaitlin Brown	52	Female	Kellyberg	South Dakota	USA
CUST1492	Darrell Miller	59	Female	Port Carol	Florida	USA
CUST1493	Amanda Powell	28	Male	South William	Alabama	USA
CUST1494	Justin Lopez	21	Male	Port Cheryl	Texas	USA
CUST1495	James Berger	23	Female	Bryanmouth	Delaware	USA
CUST1496	Christopher Ford	56	Male	North Seanfurt	Ohio	USA
CUST1497	Sarah Marsh	47	Female	Port William	New York	USA
CUST1498	Traci Gonzalez	35	Male	Snyderhurt	New Mexico	USA
CUST1499	Philip Pennington	27	Female	East Valeriview	Oregon	USA

Analysis Tools: Anova: Single Factor, Anova: Two-Factor With Replication, Anova: Two-Factor Without Replication, Correlation, Covariance, Descriptive Statistics, Exponential Smoothing, F-Test Two-Sample for Variances, Fourier Analysis, Histogram

File Home Insert Draw Page Layout Formulas Data Review View Help Table Design

Get Data From Text/CSV Recent Sources From Web Existing Connections Refresh All Workbook Links Queries & Connections Properties Sort & Filter Filter Clear Apply Text to Columns Data Validation Remove Duplicates Data Model Consolidate What-If Analysis Forecast Subtotal Outline Data Analysis

Customer_ID

Customer_ID	Name	Age	Gender	City	State	Country
CUST1000	Sheena Bonilla	55	Male	Donaldton	North Carolina	USA
CUST1001	Eugene Nelson	26	Female	Phillipsmouth	North Dakota	USA
CUST1002	Dr. Benjamin Price	29	Female	Kylebury	Alabama	USA
CUST1003	Jeremy Harris	59	Female	Thomastown	Ohio	USA
CUST1004	Caroline Collins	64	Female	Nicholasshire	California	USA
CUST1005	Christine Vaughn	24	Male	East Amandamouth	Oregon	USA
CUST1006	Amber Jackson	29	Female	Deborahborough	South Dakota	USA
CUST1007	Daniel Boyd	29	Female	West Pamelafurt	West Virginia	USA
CUST1008	Melissa Howard	53	Female	Maynardton	New Hampshire	USA
CUST1009	Ms. Maria Leach	64	Male	Melissashire	Pennsylvania	USA
CUST1010	Dominic Page	53	Male	Port Michaelburgh	Pennsylvania	USA
CUST1011	Brian Lopez	28	Female	East Krista	New York	USA
CUST1012	Kenneth Lambert	55	Female	North Kathryn	Kansas	USA
CUST1013	Nathan Hernandez	32	Female	New Becky	Montana	USA
CUST1014	Ashley Cohen	53	Female	Lake Kristamouth	Nebraska	USA
CUST1015	Laura Alvarez	49	Female	Lawsonhaven	Washington	USA
CUST1016	John Scott	24	Female	South Christopher	Pennsylvania	USA
CUST1017	Adam Cruz	60	Female	Brandonville	Georgia	USA
CUST1018	Rhonda Moss	43	Male	South William	Kansas	USA
CUST1019	Michael Torres	46	Female	Gonzalezfort	New Jersey	USA
CUST1020	Bridget Allen	59	Male	Lake Shaneburgh	Delaware	USA
CUST1021	Jeff Martinez	36	Female	Smithburgh	Oklahoma	USA
CUST1022	Paul Fox	33	Female	Michaelton	Hawaii	USA
CUST1023	Jonathan Steele	52	Female	Molinabury	Oregon	USA
CUST1024	Heather Collins	40	Male	Rogerschester	Idaho	USA
CUST1025	Diana Taylor	50	Female	Vasqueztown	California	USA
CUST1026	Amber Carlson	18	Female	East Michellemouth	Missouri	USA

Descriptive Statistics: Input Range: \$C\$2:\$C\$501, Grouped By: Columns, Output options: New Workbook, Summary statistics: Confidence Level for Mean: 95%, kth Largest: 1, kth Smallest: 1

Customer_Dim - AGE														
Metric	Mean	Standard	Median	Mode	Standard Deviation	Sample Variance	Kurtosis	Skewness	Range	Minimum	Maximum	Sum	Count	
Age	41.562	0.611385	42	46	13.67098928	186.8959479	-1.219391989	0.03654853	46	18	64	20781	500	

Product_Dim-COST and STOCK														
Metric	Mean	Standard Error	Median	Mode	Standard Deviation	Sample Variance	Kurtosis	Skewness	Range	Minimum	Maximum	Sum	Count	
Cost	255.2938	13.11680407	244.84	358.47	131.1680407	17205.0549	-0.972612663	0.024172762	478.41	17.31	495.72	25529.38	100	
Stock	273.975	13.04546544	278.5	82	130.4546544	17018.41684	-1.180380281	-0.018630837	439	57	496	27397.5	100	

Sales_Fact - QUANTITY, UNIT_PRICE, DISCOUNT, TOTAL_AMOUNT														
Metric	Mean	Standard Error	Median	Mode	Standard Deviation	Sample Variance	Kurtosis	Skewness	Range	Minimum	Maximum	Sum	Count	
Quantity	2.482	0.024852501	2	2	1.111437649	1.235293567	-1.344127634	0.022378357	3	1	4	4964	2000	
Unit_Price	513.596	6.311345168	522.43	871.34	282.3324349	79711.60381	-1.193099186	-0.034758137	979.8	20.09	999.89	1027191.96	2000	
Discount	0.14803	0.001958463	0.14	0.1	0.087585128	0.007671155	-1.707195864	0.063099261	0.3	0	0.3	296.06	2000	
Total_Amount	1087.197	18.32723712	868.7182	#N/A	819.618961	671775.2412	0.045826174	0.879047043	3889.7838	18.459	3908.7428	2174394.71	2000	

Combined_Table-GROSS_SALES,TOTAL_COST,DISCOUNT_AMOUNT,PROFIT														
Metric	Mean	Standard Error	Median	Mode	Standard Deviation	Sample Variance	Kurtosis	Skewness	Range	Minimum	Maximum	Sum	Count	
Gross_Sales	1278.223	21.27240076	1008.515	2614.02	951.330683	905030.0684	-0.14808734	0.817734912	3971.66	21.14	3992.8	2556446.85	2000	
Total_Cost	643.0339	10.3487078	525.5	716.94	462.8082826	214191.5064	-0.039334136	0.836011165	1965.57	17.31	1982.88	1286067.85	2000	
Discount_Amount	221.0376	20.76302608	119.40805	0	928.5507546	862206.5038	1407.802717	35.44212511	38137.84	0	38137.84	442075.249	2000	
Profit	444.1634	17.6123917	349.2078	#N/A	787.650102	620392.6831	0.719190383	0.595089543	5331.1536	-1705.5056	3625.648	888326.847	2000	

STEP-5

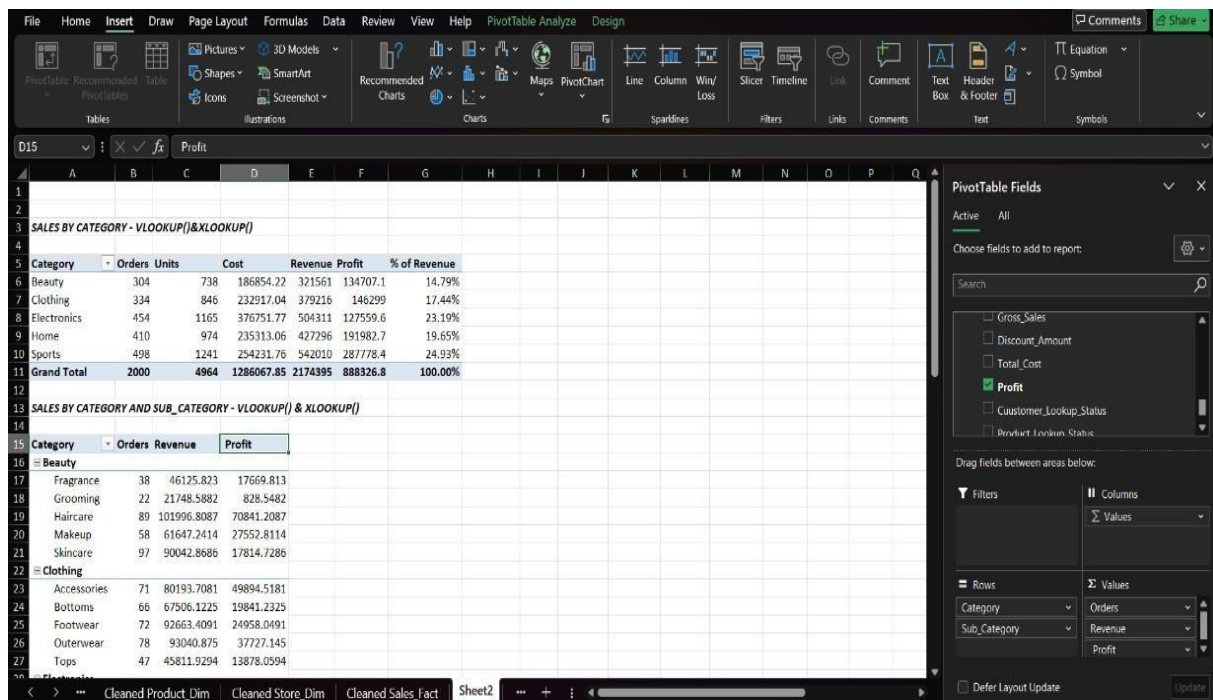
CREATING PIVOT TABLES.

Microsoft Excel interface showing a PivotTable from table or range dialog box. The dialog box prompts the user to select a table or range, choose where to place the PivotTable (New Worksheet or Existing Worksheet), and whether to analyze multiple tables. The background shows a PivotTable with columns: Sales_ID, Order_Date, Year, Month, Customer_ID, Product_ID, Store_ID, Quantity, Unit_Price, Discount, Payment_Type, Total_Amount, Customer_Name, and Age. The PivotTable is currently set to show data by Customer_ID, Product_ID, and Store_ID.

Sales_ID	Order_Date	Year	Month	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Customer_Name	Age
SALE0	07-07-2024	2024	7	CUST1409	PROD97	STORE3	1	218.04	9%	Credit Card	198.4164	Sarah Smith	55
SALE1	25-12-2023	2023	12	CUST1467	PROD74	STORE17	3	104.47	27%	PayPal	228.7893	Michael Foley	63
SALE2	03-06-2024	2024	6	CUST1056	PROD49	STORE8	4	914.05	25%	Credit Card	2742.15	Christopher Sutton	49
SALE3	29-11-2023	2023	11	CUST1346	PROD76	STORE8	1	921.51	16%	COD	774.0684	John Jordan	58
SALE4	06-01-2024	2024	1	CUST1493	PROD59	STORE5	3	745.3	18%	Credit Card	1833.438	Amanda Powell	28
SALE5	26-07-2024	2024	7	CUST1116	PROD15	STORE7	3	147.01	19%	COD	357.2343	Anthony Phillips	18
SALE6	12-07-2025	2025	7	CUST1016	PROD56	STORE4	2	785.31	15%	PayPal	1335.027	John Scott	24
SALE7	18-11-2023	2023	11	CUST1152	PROD70	STORE8	4	342.39	23%	PayPal	1054.5612	Patrick Roberts	41
SALE8	04-05-2025	2025	5	CUST1187	PROD99	STORE5	3	448.25	7%	PayPal	1250.6175	Dr. Suzanne Bullock	60
SALE9	31-07-2025	2025	7	CUST1464	PROD33	STORE0	2	873.17	23%	PayPal	1344.6818	Lindsay Beasley	45
SALE10	13-02-2024	2024	2	CUST1065	PROD24	STORE4	3	320.68	21%	Credit Card	259.1259	Michael Thomas	21
SALE11	24-05-2025	2025	5	CUST1320	PROD39	STORE4	3	320.68	21%	Credit Card	952.5744	Jessica Gonzalez	34
SALE12	24-08-2024	2024	8	CUST1237	PROD88	STORE4	3	320.68	21%	Credit Card	2552.5368	Wesley Short	41
SALE13	06-08-2025	2025	8	CUST1443	PROD34	STORE4	3	320.68	21%	PayPal	443.66	Adam Castro	51
SALE14	21-06-2025	2025	6	CUST1348	PROD24	STORE4	3	320.68	21%	PayPal	1145.8548	Sherry Cabrera	55
SALE15	10-12-2024	2024	12	CUST1435	PROD36	STORE4	3	320.68	21%	COD	2509.4592	Lindsay Beasley	37
SALE16	13-09-2025	2025	9	CUST1276	PROD65	STORE4	3	320.68	21%	Credit Card	677.97	Arthur Warren	38
SALE17	11-12-2024	2024	12	CUST1079	PROD27	STORE4	3	320.68	21%	Credit Card	774.1426	Molly Gardner	22
SALE18	13-08-2024	2024	8	CUST1094	PROD39	STORE4	3	320.68	21%	COD	838.845	David Patterson	43
SALE19	18-03-2024	2024	3	CUST1239	PROD60	STORE4	3	320.68	21%	COD	1782.4512	Maria Hill	58
SALE20	05-07-2025	2025	7	CUST1144	PROD72	STORE4	3	320.68	21%	COD	2159.4276	Justin Cross	62
SALE21	04-06-2024	2024	6	CUST1396	PROD43	STORE4	3	320.68	21%	Credit Card	459.8568	Alexander Byrd	25
SALE22	12-04-2024	2024	4	CUST1476	PROD69	STORE4	3	320.68	21%	PayPal	2850.1122	Elizabeth Foster	33
SALE23	23-06-2024	2024	6	CUST1368	PROD12	STORE4	3	320.68	21%	Credit Card	661.7376	Miranda King	34
SALE24	08-09-2025	2025	9	CUST1186	PROD67	STORE4	3	320.68	21%	Credit Card	216.504	Jennifer Pruitt	53
SALE25	17-09-2024	2024	9	CUST1381	PROD5	STORE3	4	852.71	14%	PayPal	2933.3224	Erika Padilla	50
SALE26	03-01-2024	2024	1	CUST1213	PROD98	STORE8	4	156.16	400%	Credit Card	599.6544	Stuart Johnson	42

Microsoft Excel interface showing the PivotTable Fields task pane. The task pane displays the PivotTable Name (PivotTable2) and the Active Field (Sales_ID). The task pane also shows the PivotTable Fields list, including Sales_ID, Order_Date, and Year. The task pane includes a search bar and a list of fields to add to the report. The background shows a PivotTable with columns: Sales_ID, Order_Date, Year, Month, Customer_ID, Product_ID, Store_ID, Quantity, Unit_Price, Discount, Payment_Type, Total_Amount, Customer_Name, and Age. The PivotTable is currently set to show data by Customer_ID, Product_ID, and Store_ID.

Sales_ID	Order_Date	Year	Month	Customer_ID	Product_ID	Store_ID	Quantity	Unit_Price	Discount	Payment_Type	Total_Amount	Customer_Name	Age
SALE0	07-07-2024	2024	7	CUST1409	PROD97	STORE3	1	218.04	9%	Credit Card	198.4164	Sarah Smith	55
SALE1	25-12-2023	2023	12	CUST1467	PROD74	STORE17	3	104.47	27%	PayPal	228.7893	Michael Foley	63
SALE2	03-06-2024	2024	6	CUST1056	PROD49	STORE8	4	914.05	25%	Credit Card	2742.15	Christopher Sutton	49
SALE3	29-11-2023	2023	11	CUST1346	PROD76	STORE8	1	921.51	16%	COD	774.0684	John Jordan	58
SALE4	06-01-2024	2024	1	CUST1493	PROD59	STORE5	3	745.3	18%	Credit Card	1833.438	Amanda Powell	28
SALE5	26-07-2024	2024	7	CUST1116	PROD15	STORE7	3	147.01	19%	COD	357.2343	Anthony Phillips	18
SALE6	12-07-2025	2025	7	CUST1016	PROD56	STORE4	2	785.31	15%	PayPal	1335.027	John Scott	24
SALE7	18-11-2023	2023	11	CUST1152	PROD70	STORE8	4	342.39	23%	PayPal	1054.5612	Patrick Roberts	41
SALE8	04-05-2025	2025	5	CUST1187	PROD99	STORE5	3	448.25	7%	PayPal	1250.6175	Dr. Suzanne Bullock	60
SALE9	31-07-2025	2025	7	CUST1464	PROD33	STORE0	2	873.17	23%	PayPal	1344.6818	Lindsay Beasley	45
SALE10	13-02-2024	2024	2	CUST1065	PROD24	STORE4	3	320.68	21%	Credit Card	259.1259	Michael Thomas	21
SALE11	24-05-2025	2025	5	CUST1320	PROD39	STORE4	3	320.68	21%	Credit Card	952.5744	Jessica Gonzalez	34
SALE12	24-08-2024	2024	8	CUST1237	PROD88	STORE4	3	320.68	21%	Credit Card	2552.5368	Wesley Short	41
SALE13	06-08-2025	2025	8	CUST1443	PROD34	STORE4	3	320.68	21%	PayPal	443.66	Adam Castro	51
SALE14	21-06-2025	2025	6	CUST1348	PROD24	STORE4	3	320.68	21%	PayPal	1145.8548	Sherry Cabrera	55
SALE15	10-12-2024	2024	12	CUST1435	PROD36	STORE4	3	320.68	21%	COD	2509.4592	Lindsay Beasley	37
SALE16	13-09-2025	2025	9	CUST1276	PROD65	STORE4	3	320.68	21%	Credit Card	677.97	Arthur Warren	38
SALE17	11-12-2024	2024	12	CUST1079	PROD27	STORE4	3	320.68	21%	Credit Card	774.1426	Molly Gardner	22
SALE18	13-08-2024	2024	8	CUST1094	PROD39	STORE4	3	320.68	21%	COD	838.845	David Patterson	43
SALE19	18-03-2024	2024	3	CUST1239	PROD60	STORE4	3	320.68	21%	COD	1782.4512	Maria Hill	58
SALE20	05-07-2025	2025	7	CUST1144	PROD72	STORE4	3	320.68	21%	COD	2159.4276	Justin Cross	62
SALE21	04-06-2024	2024	6	CUST1396	PROD43	STORE4	3	320.68	21%	Credit Card	459.8568	Alexander Byrd	25
SALE22	12-04-2024	2024	4	CUST1476	PROD69	STORE4	3	320.68	21%	PayPal	2850.1122	Elizabeth Foster	33
SALE23	23-06-2024	2024	6	CUST1368	PROD12	STORE4	3	320.68	21%	Credit Card	661.7376	Miranda King	34
SALE24	08-09-2025	2025	9	CUST1186	PROD67	STORE4	3	320.68	21%	Credit Card	216.504	Jennifer Pruitt	53
SALE25	17-09-2024	2024	9	CUST1381	PROD5	STORE3	4	852.71	14%	PayPal	2933.3224	Erika Padilla	50
SALE26	03-01-2024	2024	1	CUST1213	PROD98	STORE8	4	156.16	400%	Credit Card	599.6544	Stuart Johnson	42

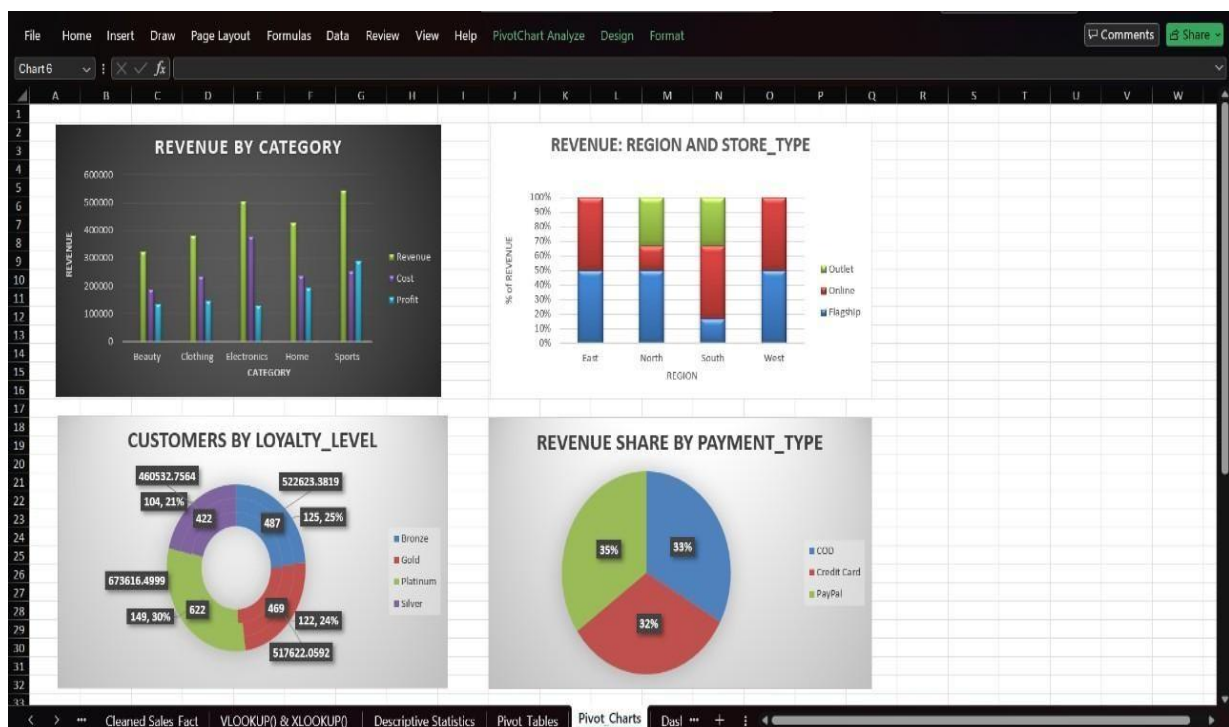


Here, 14 pivot tables are Created .

STEP-6

CREATING PIVOT CHARTS.

In the PIVOT_TABLE sheet, there is the “PIVOT TABLE ANALYZE” ribbon. Goto the TOOLS group , select PIVOT CHARTS. Then choose the appropriate chart and click OK.



Here, 10 pivot charts are created.

STEP_7

DATA VISUALIZATION

The Interactive Dashboard is created and the slicers are added.

